

Foreground-Aware Zero-Shot Plant Trait Segmentation in Trellised Crops Using Side View Monocular RGB Images

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Abstract. Zero-shot plant phenotyping in trellised crops is challenging because side-view images contain visually similar vegetation rows, causing foundation models to mix foreground and background organs; tiled high-resolution inference also fragments masks. We propose a geometry-aware, training-free framework combining monocular depth and open-vocabulary segmentation. Depth Anything V2 and *Valley-Aware Depth Thresholding* (VADT) isolate the closest row, while overlapping SAM3 tiles are reconstructed by *Graph-Based Mask Assembly* (GMA) and fused with the foreground mask. We evaluate 158 RGB images with approximately 10,000 annotated instances from grapevine, apple, and apricot, covering leaves, fruits, black rot, and downy mildew. VADT reaches 0.85 Dice for foreground extraction, versus 0.55 for the strongest RGB-only baseline. On vineyard images, the performance of segmentation based on full pipeline obtains in average Dice scores of 0.95 for leaves, 0.92 for fruits, 0.61 for black rot, and 0.73 for downy mildew, outperforming recent zero-shot methods. These results show that monocular geometry is an effective prior for phenotyping structured crop scenes.

Keywords: Plant Phenotyping · Zero-Shot Segmentation · Foundation Models · Monocular Depth Estimation · Trellised Crops

1 Introduction

Image-based phenotyping supports canopy characterization, yield estimation, and disease assessment. In vineyards and modern orchards, side-view images often contain several rows with nearly identical leaves, fruits, and symptoms, causing semantic models to mix foreground and background vegetation. Tiled inference helps recover small berries and lesions in high-resolution images, but fragments instances at tile boundaries.

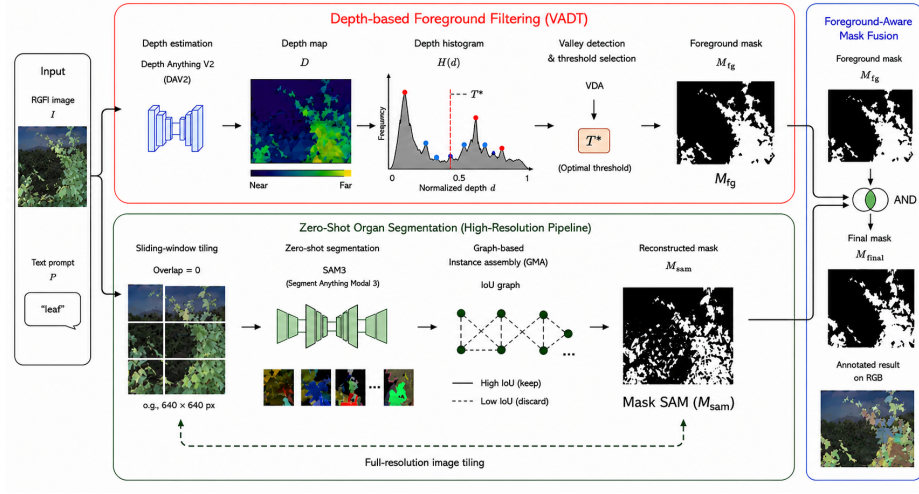


Fig. 1: Overview of the proposed training-free framework.

Most plant segmentation methods remain supervised and crop-specific, while zero-shot pipelines rarely exploit trellised-scene geometry [4, 6, 8]. Vision foundation models provide prompt-driven segmentation [2, 5, 12], whereas monocular depth models [13] recover relative geometry from a single RGB image. We combine both capabilities to isolate the target row before trait segmentation.

Successive crop rows form distinct modes in the depth distribution. VADT selects the low-density valley separating the closest row; tiled SAM3 inference, GMA, and geometric fusion then recover the requested trait. The contributions are: automatic foreground-row extraction, a geometry-aware pipeline requiring neither task-specific training nor a depth sensor, and evaluation across three crops and four phenotypic traits.

2 Geometry-Aware Zero-Shot Segmentation

Given an RGB image I and a text prompt q , the objective is to segment target instances belonging only to the closest vegetation row. Figure 1 summarizes the framework.

Valley-Aware Depth Thresholding. Depth Anything V2 predicts a relative depth map D [13]. Its normalized histogram $H(x)$, $x \in [0, 255]$, is Gaussian-smoothed into $H_s(x)$. Prominent maxima form $P = \{p_0, \dots, p_n\}$; the first is discarded because it usually represents invalid values or distant background. Valleys are computed between consecutive remaining modes, and the deepest among the first four candidates defines the threshold:

$$v_i = \arg \min_{x \in [p_i, p_{i+1}]} H_s(x), \quad T^* = \arg \min_{v \in V} H_s(v). \quad (1)$$

Table 1: Comparison with state-of-the-art segmentation methods under a unified sliding-window protocol on the vineyard subset using Dice metric. *Compared methods: Grounding DINO + SAM2 (GD-SAM2) [10, 11]; Florence-2 + SAM2 (Fl-SAM2) [10, 12] and INSID3 [3].*

Target	GD-SAM2	Fl-SAM2	INSID3	Ours
Leaves	0.84 ± 0.04	0.81 ± 0.05	0.90 ± 0.04	0.95 ± 0.02
Fruits	0.41 ± 0.02	0.13 ± 0.08	0.58 ± 0.20	0.92 ± 0.06
Black rot	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.61 ± 0.11
Downy mildew	0.14 ± 0.09	0.12 ± 0.07	0.16 ± 0.10	0.73 ± 0.14

Thresholding D at T^* yields the foreground mask $\mathcal{M}_{\text{fg}}$. VADT uses only relative depth ordering and therefore requires no metric calibration.

High-resolution inference, assembly, and fusion. The image is divided into overlapping 640×640 tiles with 50% overlap and segmented by SAM3 [2]. Fragments below 50 pixels or covering more than 55% of a tile are removed. In GMA, fragments are graph nodes; pairs are linked when bounding-box IoU exceeds 0.05 and mask IoU exceeds 0.10. Connected components reconstruct instances, while confidence scores resolve overlapping pixels. The semantic mask is finally filtered by

$$\mathcal{M}_{\text{final}} = \mathcal{M}_{\text{SAM}} \cap \mathcal{M}_{\text{fg}}, \quad (2)$$

which suppresses detections from neighboring rows.

3 Experimental Protocol and Results

Dataset and baselines. The benchmark contains 158 RGB images: 88 grapevine, 35 apricot, and 35 apple images acquired at six locations with varied cameras, viewpoints, illumination, and field conditions. Apple images come from the DeepPhenoTree-CVPPA challenge [7]. Approximately 10,000 instances were annotated at pixel level. Leaves and fruits are evaluated for all crops. Black rot and downy mildew are evaluated on grapevine. Parameters were fixed on an independent validation set. VADT is compared with BiRefNet [14], IS-Net [9], and BRIA RMBG-1.4 [1]. The full pipeline is compared with Grounding DINO+SAM2, Florence-2+SAM2, and INSID3 [3, 10–12] under the same sliding-window protocol.

Results. VADT obtains 0.85 Dice and 0.77 IoU for foreground extraction, versus 0.55 Dice and 0.47 IoU for BRIA RMBG-1.4. The strongest RGB-only baseline BiRefNet and IS-Net nearly fail. Relative depth is therefore more discriminative than appearance when successive rows share similar colors and textures.

Table 1 shows the best Dice for every vineyard target, with the largest gains for fruits and disease symptoms in average. Across species, leaf Dice reaches 0.95 for grapevine, 0.87 for apple, and 0.77 for apricot; fruit Dice reaches 0.92, 0.88,

and 0.91, respectively. Specific prompts, such as *grapes* or descriptions combining symptom color and morphology, outperform generic terms, although no weights or benchmark-specific parameters are tuned.

Ablations confirm that the modules are complementary: tiling improves small-target detection, foreground filtering removes neighboring rows, and GMA reconstructs masks across tile boundaries more consistently than non-maximum suppression. The main limitation is the global depth histogram, which can be weakened by sparse canopies or distorted by protruding branches; vegetation-weighted histograms and local depth-outlier filtering are promising extensions.

4 Conclusion

We presented a geometry-aware, training-free framework for zero-shot plant trait segmentation from side-view monocular RGB images. VADT isolates the closest crop row, GMA reconstructs tiled predictions, and geometric fusion removes neighboring vegetation. Results across grapevine, apple, and apricot show that monocular depth effectively complements semantic foundation models without crop-specific training or dedicated depth hardware.

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Software availability. To make the proposed approach accessible and facilitate its application and reproducibility, we developed **FrontVegV2**, a napari plugin implementing the presented methodology. The plugin is freely available as open-source software through Napari-hub: <https://napari-hub.org/plugins/frontveg2.html>

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